Analysis of Massive Data Sets

http://www.fer.hr/predmet/avsp

Izv. prof. dr. sc. Marin Šilić

Izv. prof. dr. sc. Goran Delač

Izv. prof. dr. sc. Klemo Vladimir

Doc. dr. sc. Adrian Satja Kurdija

Faculty of Electrical Engineering and Computing
Consumer Computing Laboratory

Course Content

- **What are the expected outcomes of the course?**
  - Deal with different types of data
    - High dimensional data
    - Graphs
    - Infinite/never-ending data
    - Labeled data
  - Use different computation models
    - MapReduce
    - Streams and online algorithms
    - In-memory single machine

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Course Content

- **What are the expected outcomes of the course?**
  - Solve real-world problems
    - Recommender Systems
    - Market Basket Analysis
    - Spam filters
    - Duplicate document detection
  - Gain various skills spanning through different areas
    - Linear Algebra (SVD, Rec. Sys., Detecting Communities)
    - Optimization (stochastic gradient descent)
    - Dynamic Programming (Frequent itemsets)
    - Hashing (LSH, Bloom filters)

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Course Content

- **Prerequisites**
  - Algorithms
    - Dynamic programming, basic data structures
  - Basic probability
    - Moments, typical distributions, MLE, …
  - Programming
    - You can do programming assignments in any language we support (Python, Java, C, C++, C# it is your choice)
    - We recommend Python, Java and C#

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Course Content

- **Lectures schedule (1. cycle)**

1. Detecting Near Duplicates – SimHash
2. Detecting Near Duplicates – MinHash
3. Recommendation Systems CF
4. Link Analysis
5. Finding Frequent Itemsets
6. Clustering
7. Communities Detection in SNGs

- Midterm – two weeks break

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Course Content

- **Lectures schedule (2. cycle)**
  8. Mining Data Streams
  9. Mining Data Streams 2
  10. Recommendation Systems MF
  11. Advertising on the Web
  12. Dimensionality Reduction

  Final exam

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Course Organization

Course stuff

- Lecturers
  - Associate professor Marin Šilić, PhD
  - Associate professor Goran Delač, PhD
  - Associate professor Klemo Vladimir, PhD
  - Assistant professor Adrian Satja Kurdija, PhD
- Teaching Assistants
  - Stjepan Požgaj, MSc
- Reference text
  - Leskovec-Rajaraman-Ullman: Mining of Massive Datasets
  - http://www.mmds.org/
- Questions
  - avsp@fer.hr

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Course Organization

- ECTS points **4**
- Lecture hours **45**
- Programming assignments **15**
- Course duration in weeks **13**

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Course Organization

- **Programming assignments**
  - 6 programming assignments
  - Students submit their solutions independently
  - The solutions will be evaluated using online judge system
  - Follow the news on the course website, instructions for the first assignment will be posted soon!

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Course Organization

Grades

- **Theory** — **70 points**
  - Midterm — 30 points
  - Final exam — 30 points
  - Lectures attendance — 10 points
- **Programming assignments** — **30 points**
  - Each assignment will be graded uniformly

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Course Organization

- **The right to take an exam**
  - Programming assignments threshold — 15 points (50%)
- **To get a positive grade**
  - Theory threshold — 35 points (50%)
  - Programming assignments threshold — 15 points (50%)

- **Grades** (*)
  - excellent (5) — ≥ 88 points
  - very good (4) — ≥ 75 points
  - good (3) — ≥ 63 points
  - sufficient (2) — ≥ 50 points

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